

Employing 0-1 test for chaos to characterize the chaotic dynamics of a generalized Gauss iterated map

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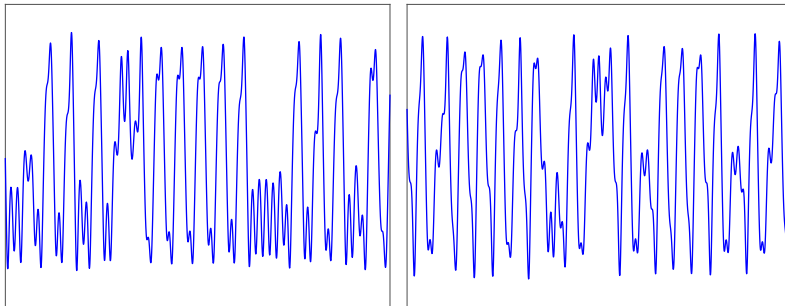


- 1 Introduction
- 2 Dynamical System
- 3 Detection of chaos: 0-1 test
- 4 Simulation results
- 5 Final Remarks

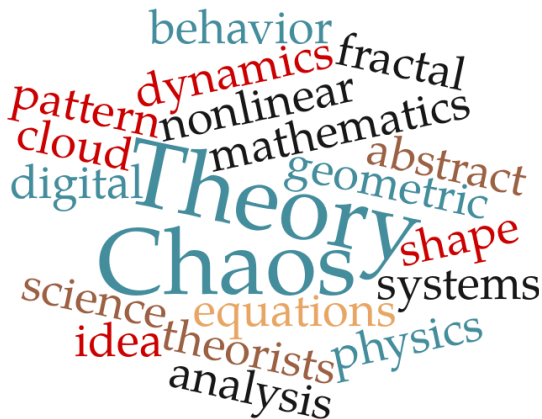
Section 1

Introduction

What's common in these time series?

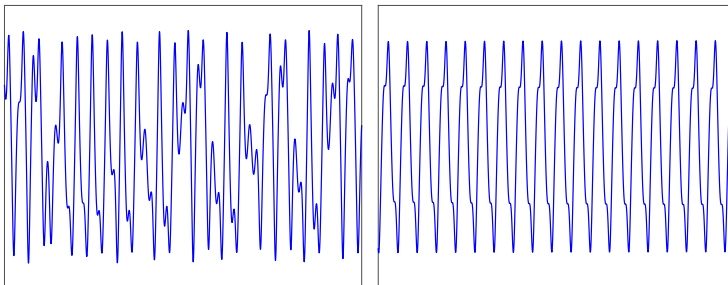


The presence of chaos!

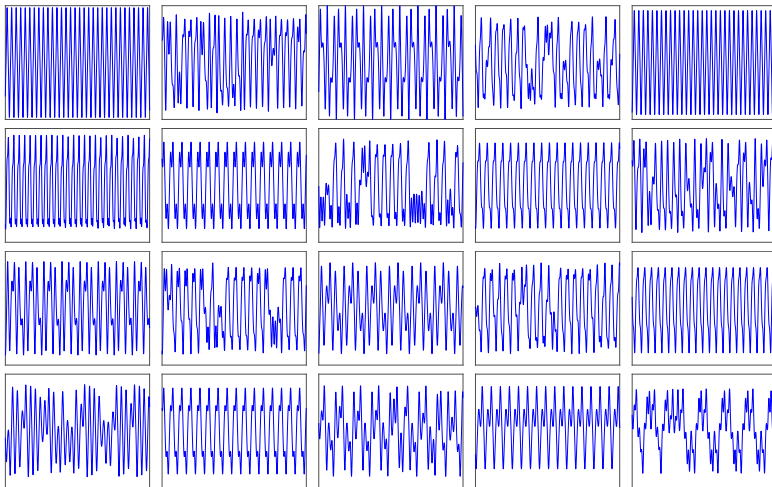


Chaos detection

- Chaos is undesirable in several applications
- It can be identified by some notion of disorder or irregularity
- Can you point out which of these time series is chaotic?



Can you point out which of these time series is chaotic?



Classification of time series

- Simple changes to a system equations or parameters can modify all the dynamic behaviour
- Identifying chaotic responses can be **expensive** if you do not have much knowledge about the system
- Lyapunov exponents are widely used in the literature
- Unfortunately the computation cost of this method is high
- The 0-1 test for chaos brings a good compromise between accuracy and low computational cost

Research objectives

This research has the following objectives:

- Investigate in detail the dynamics of a generalized Gauss map
 - Time series
 - Bifurcation diagrams
 - Test 0-1 for chaos
- Reveal some new features that did not exist in the original map
- Identify parametric regions of chaos
 - Bifurcation diagrams
 - Contour maps

Section 2

Dynamical System

Gauss iterated map

$$x_{n+1} = \exp(-\alpha x_n^2) + \beta$$

$$x_0 = a$$

- α - width parameter
- β - offset parameter
- n - number of iterations
- x_n - n-th iteration
- a - initial condition

Generalized Gauss iterated map

$$x_{n+1} = \gamma \exp\left(-\alpha(x_n - \delta)^2\right) + \beta$$

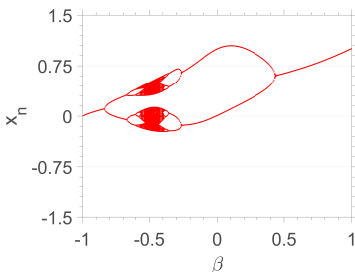
$$x_0 = a$$

- α - width parameter
- β - offset parameter
- γ - scale parameter
- δ - mean parameter
- n - number of iterations
- x_n - n-th iteration
- a - initial condition

Maps comparative

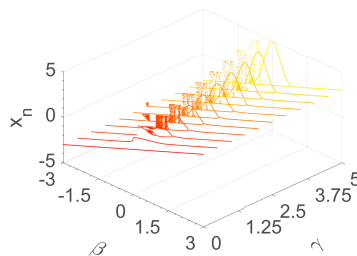
Original Gauss iterated map

- Well studied in the literature
- Manifest chaos for several values of $\beta \in [-1, 1]$, when $\alpha = 4.9$ or $\alpha = 6.2$



Generalized Gauss iterated map

- Not studied in the literature (best of our knowledge)
- Regions of chaos are unknown

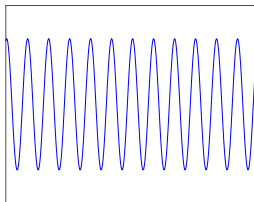


Section 3

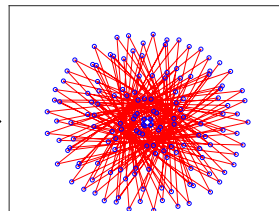
Detection of chaos: 0-1 test

Idea of the 0-1 test

regular dynamics

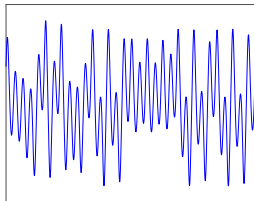


bounded dynamics

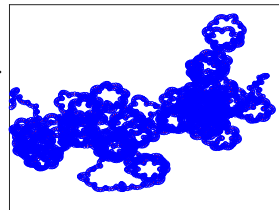


Euclidian
extension

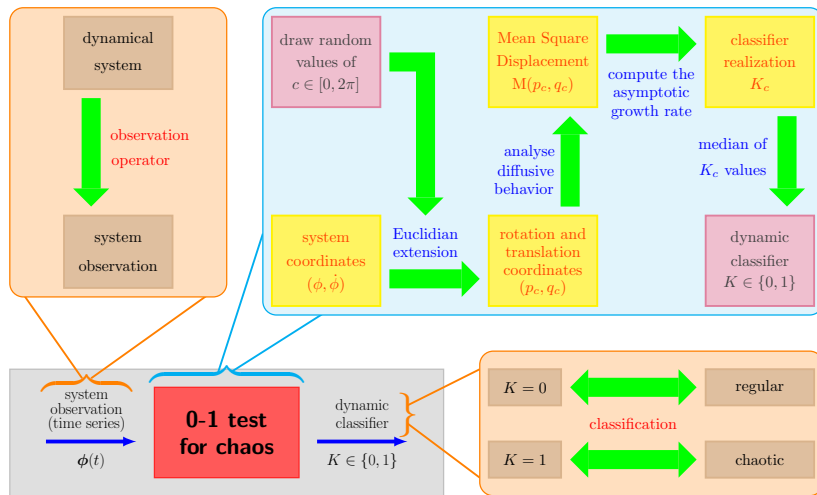
chaotic dynamics



diffusive dynamics



0-1 test for chaos schematic



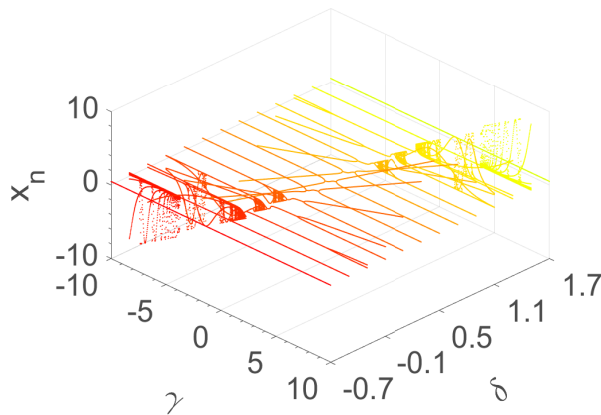
Section 4

Simulation results

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

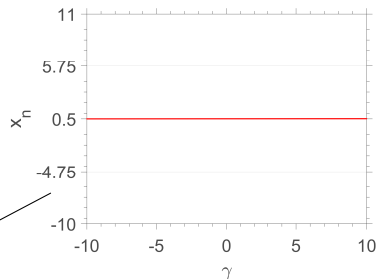
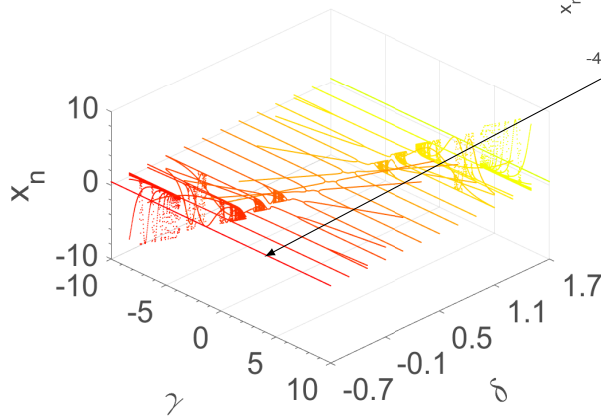
$$x_0 = 0.2$$



Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

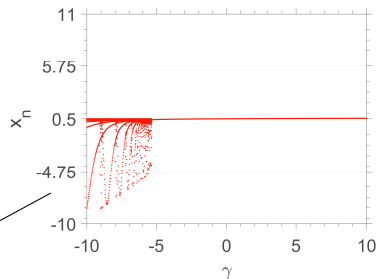
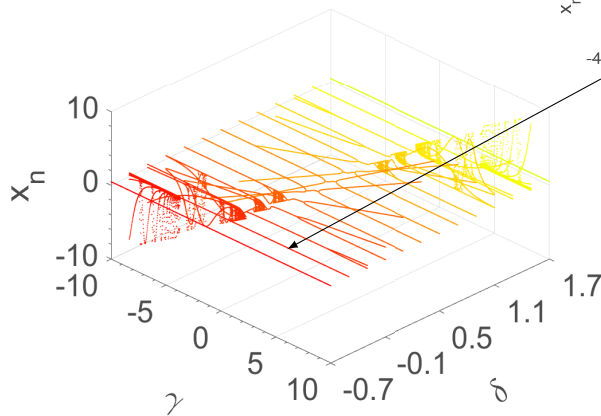


$$\delta = -0.7$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

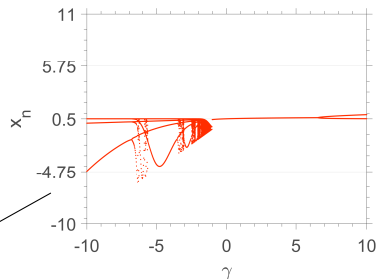
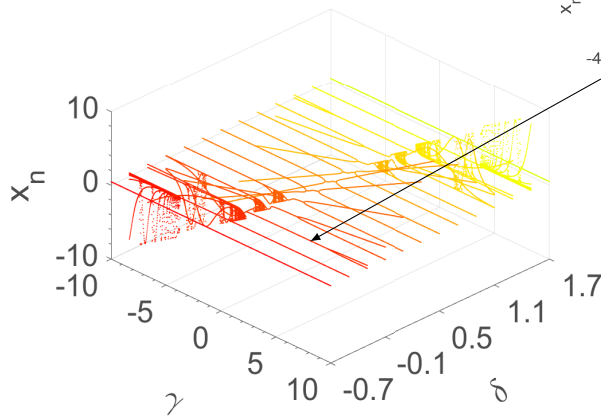


$$\delta = -0.5$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

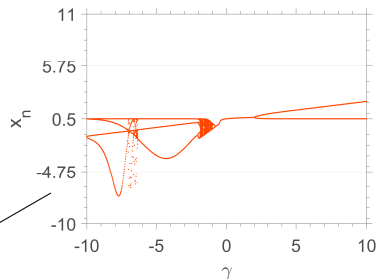
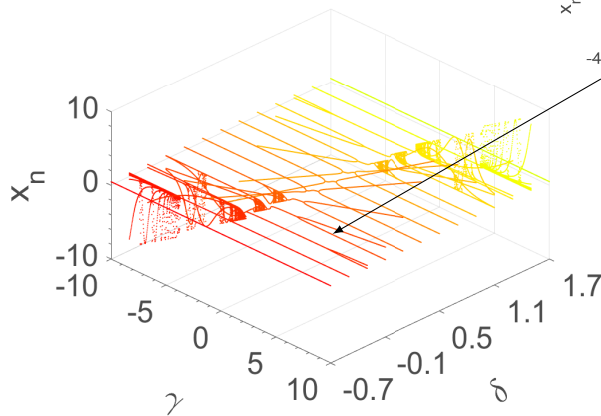


$$\delta = -0.3$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

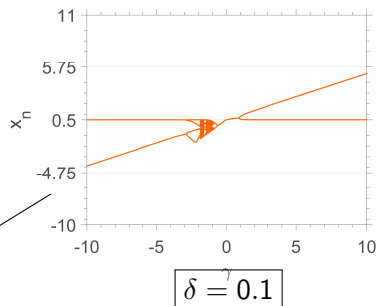
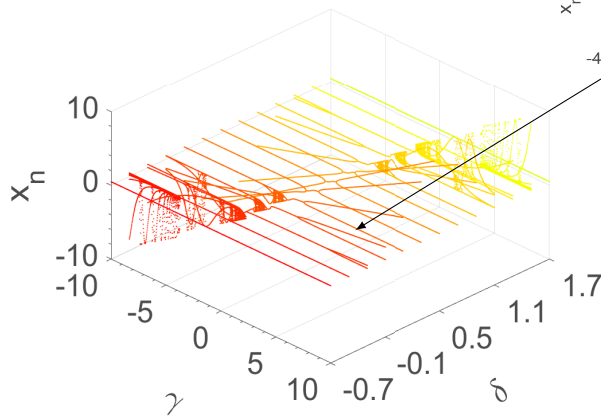


$$\delta = -0.1$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

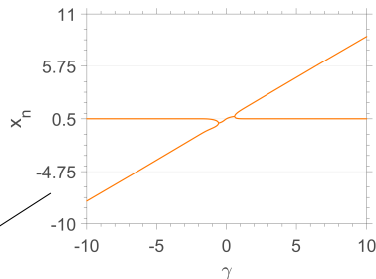
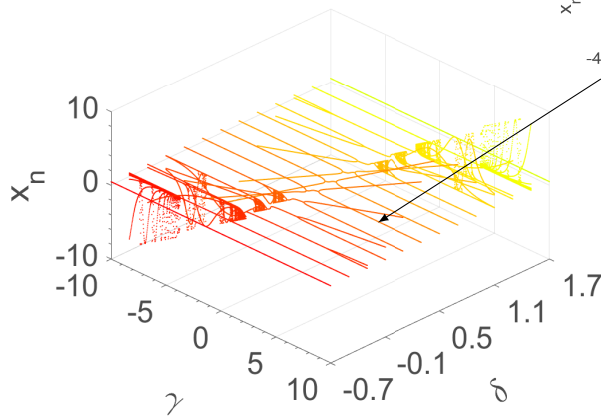
$$x_0 = 0.2$$



Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

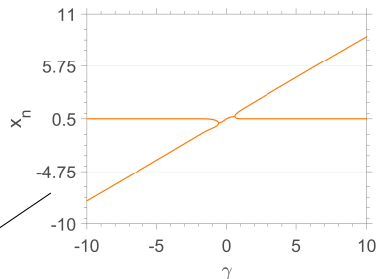
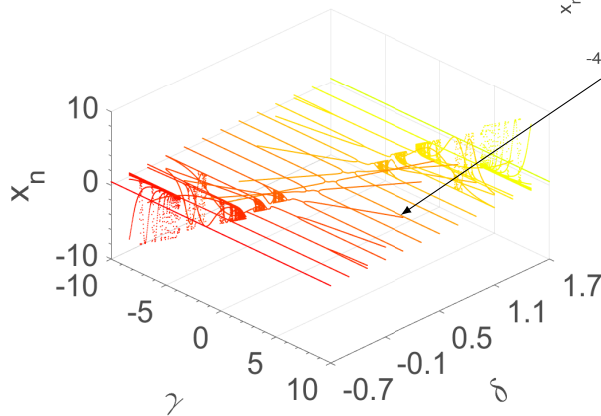


$$\delta = 0.3$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

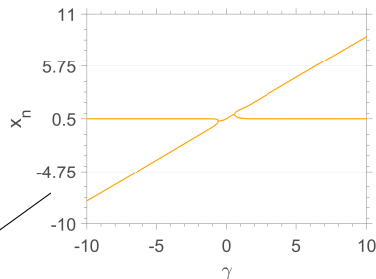
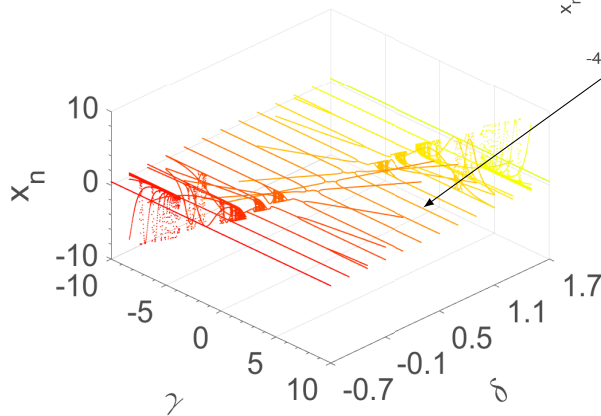


$$\delta = 0.5$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

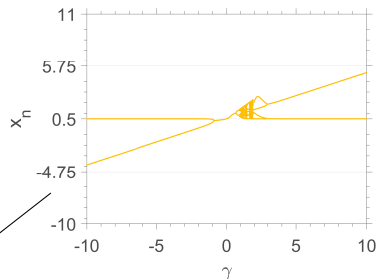
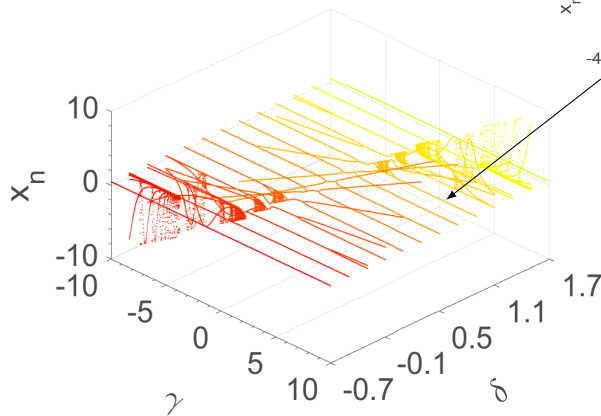


$$\delta = 0.7$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

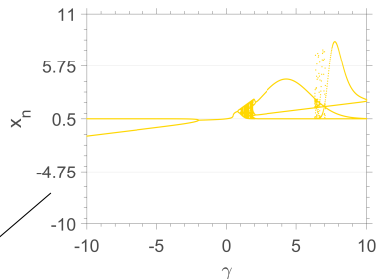
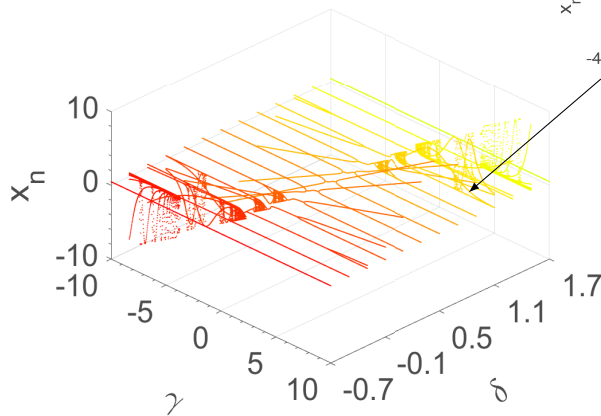


$$\delta = 0.9$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

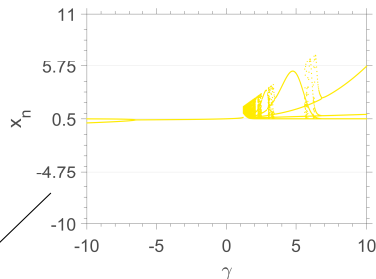
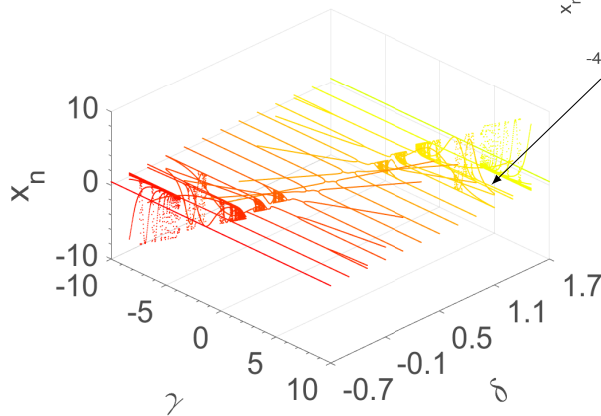


$$\delta = 1.1$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

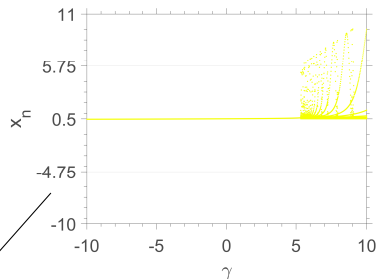
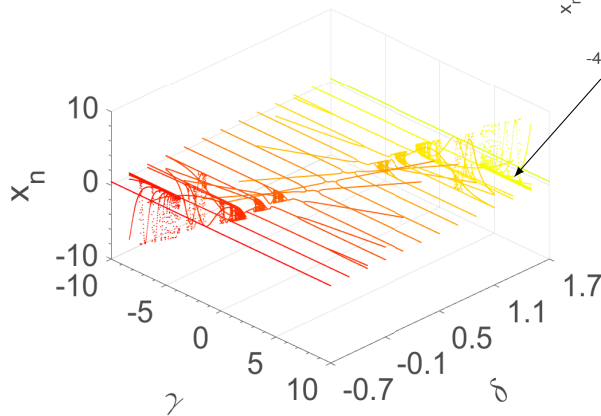


$$\delta = 1.3$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

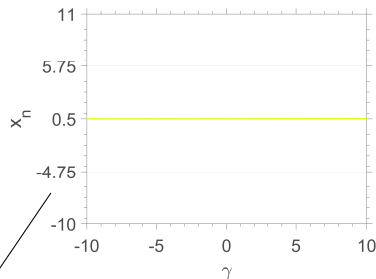
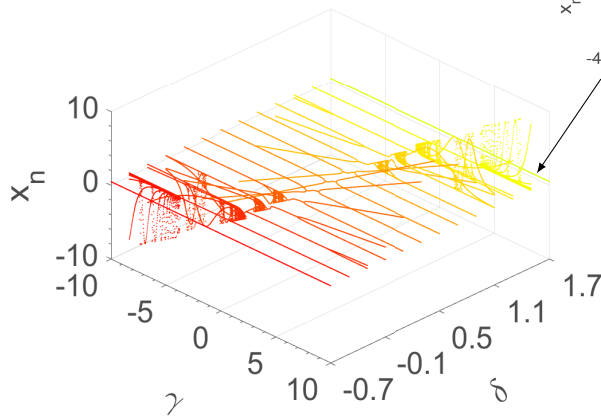


$$\delta = 1.5$$

Bifurcation diagrams: population $\times$ scale parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

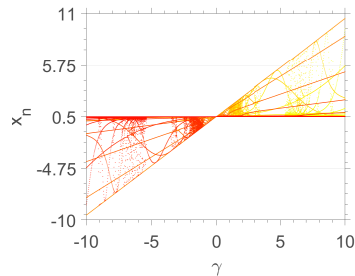
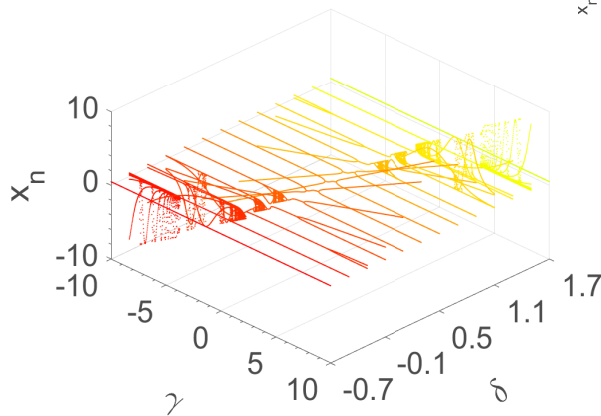


$$\delta = 1.7$$

Bifurcation diagrams: population $\times$ scale parameter

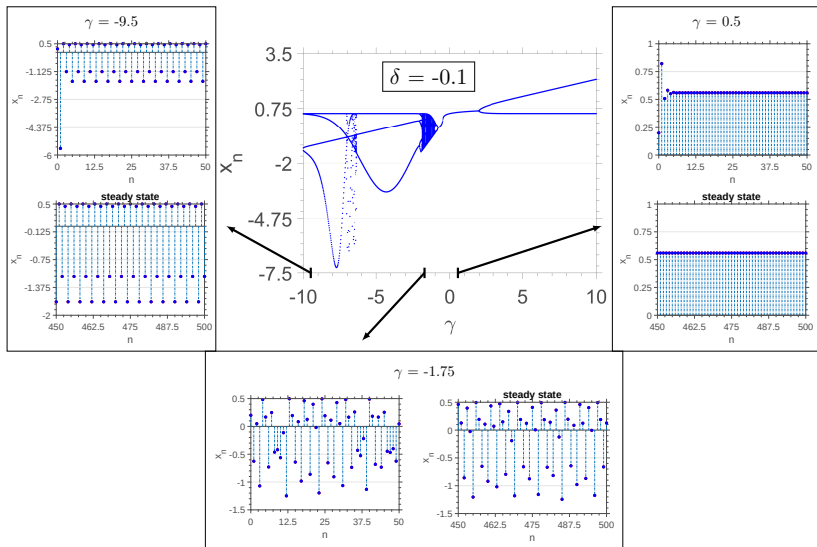
$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

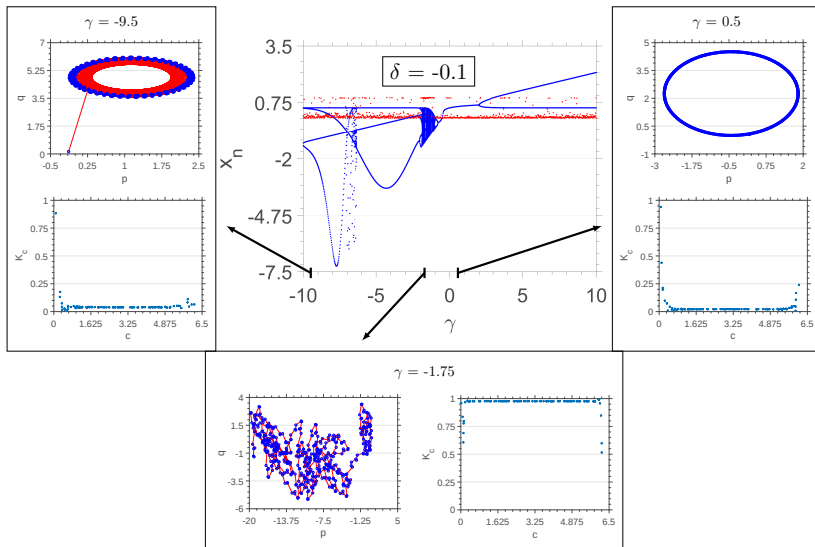


$$\delta = -0.7:0.2:1.7$$

Bifurcation diagram: population $\times$ scale parameter



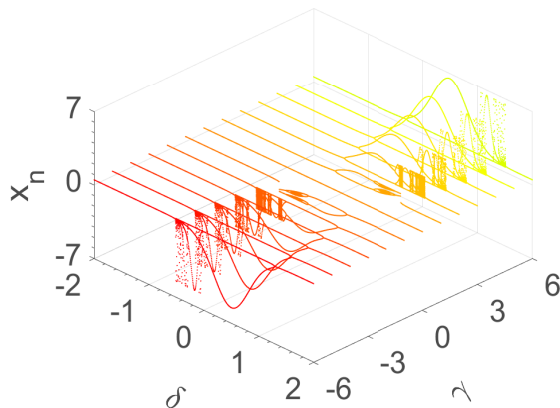
Bifurcation diagram: population $\times$ scale parameter



Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

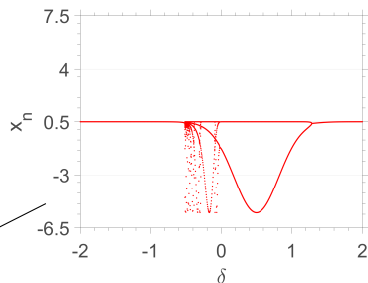
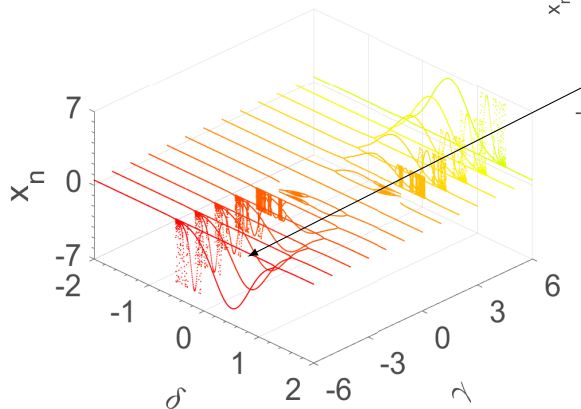
$$x_0 = 0.2$$



Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

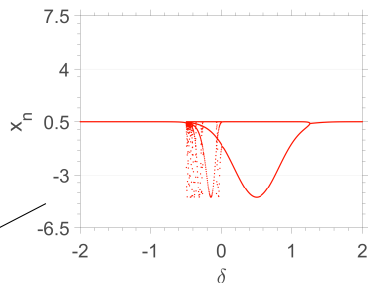
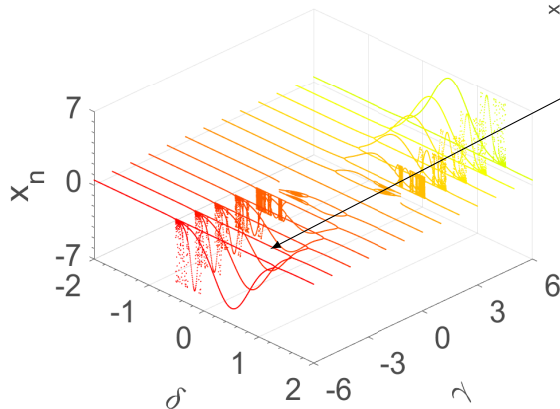


$$\gamma = -6$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

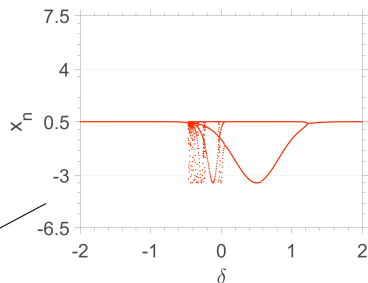
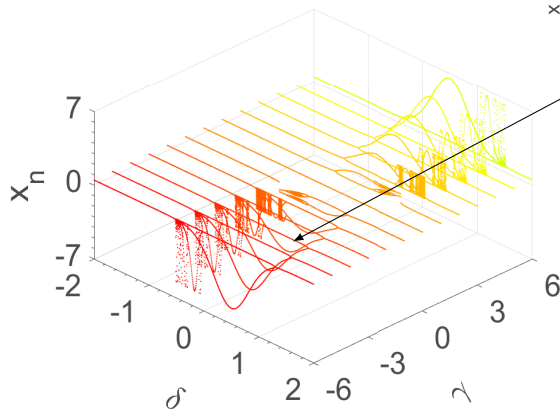


$$\gamma = -5$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

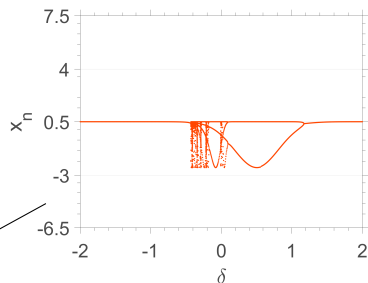
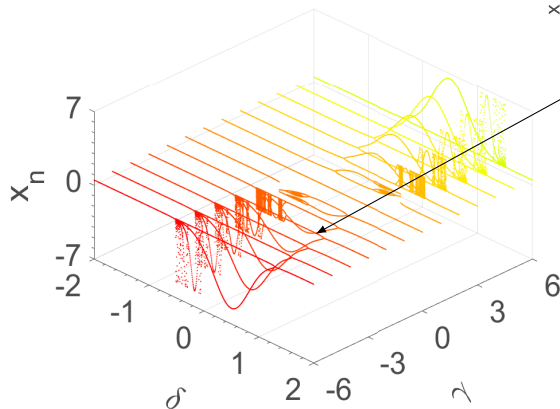


$$\gamma = -4$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

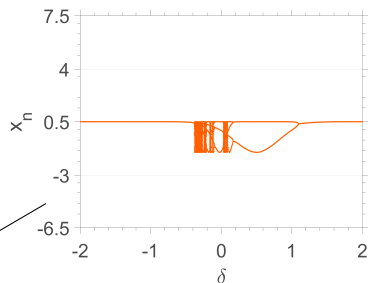
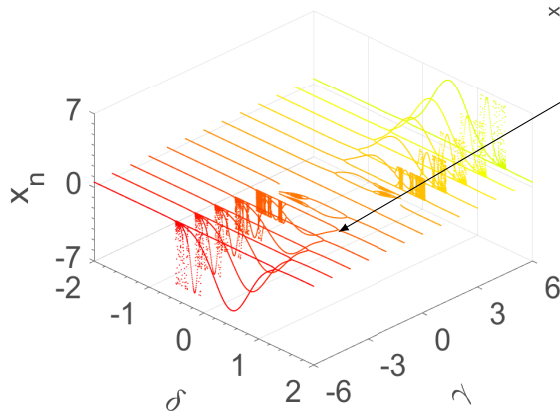


$$\gamma = -3$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

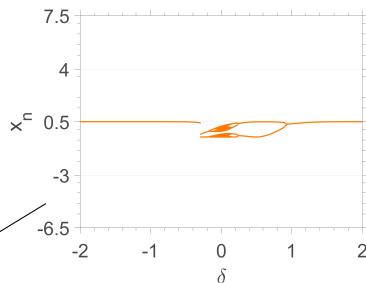
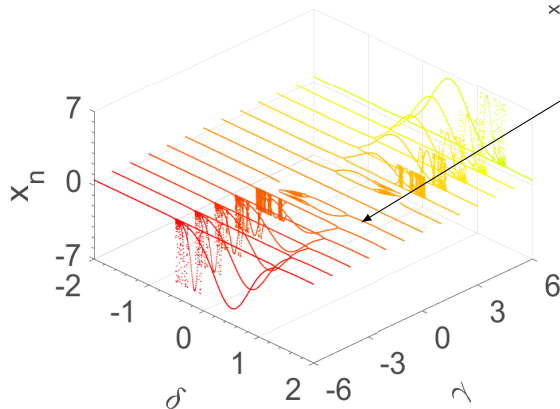


$$\gamma = -2$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

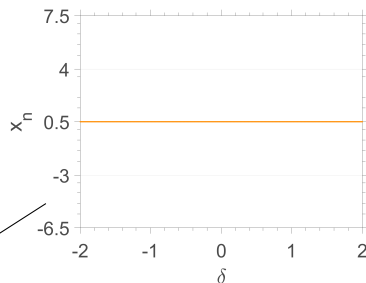
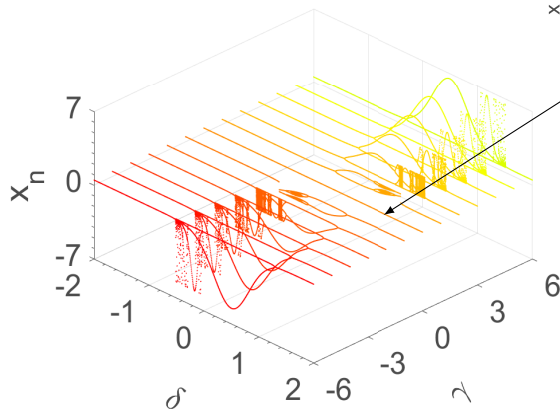


$$\gamma = -1$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

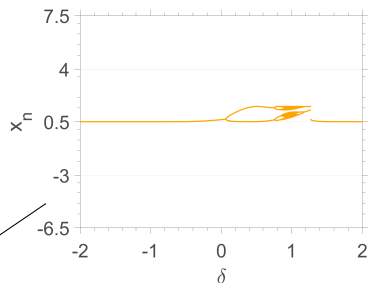
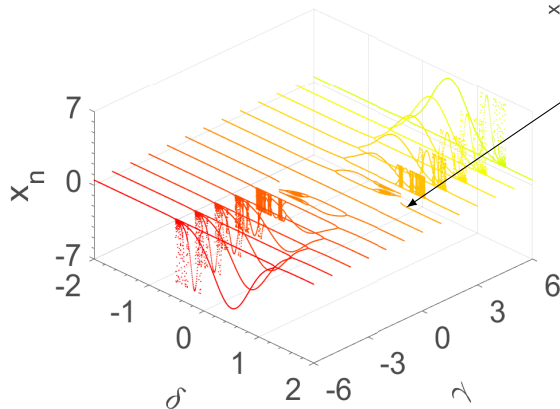


$$\gamma = 0$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

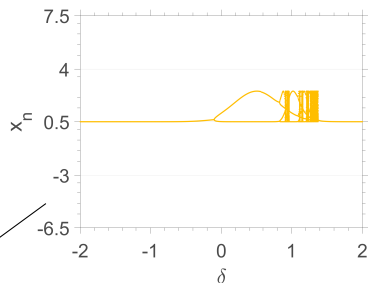
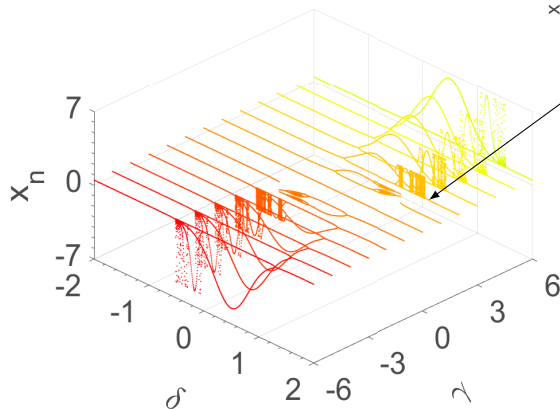


$$\gamma = 1$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

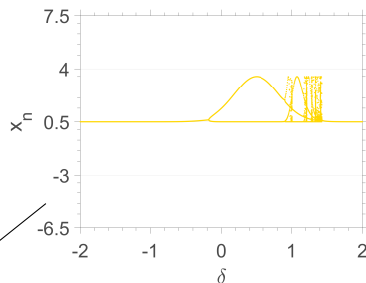
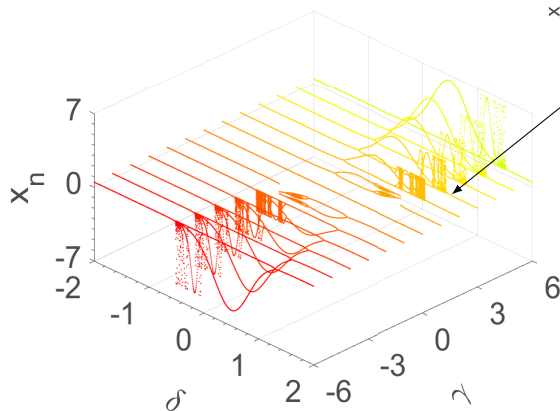


$$\gamma = 2$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

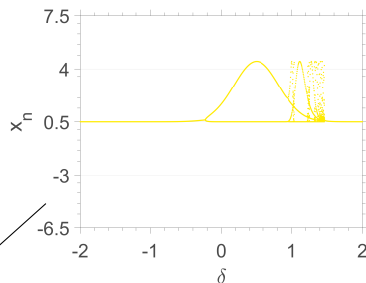
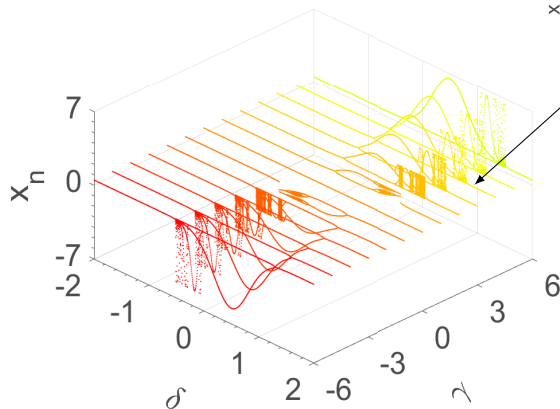


$$\gamma = 3$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

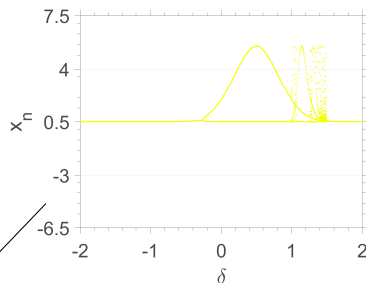
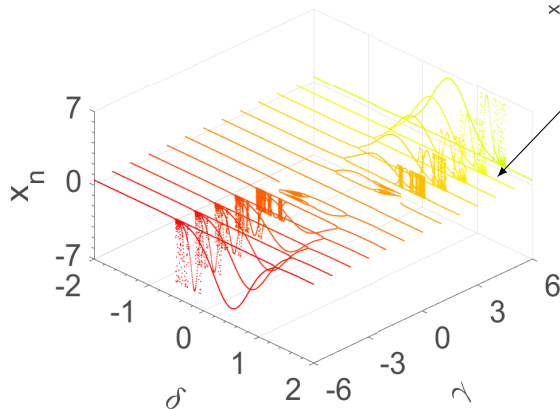


$$\gamma = 4$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

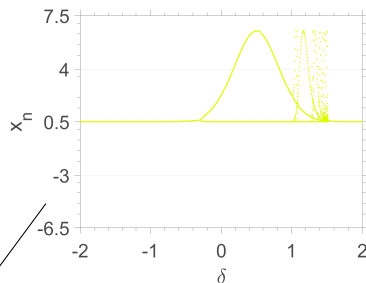
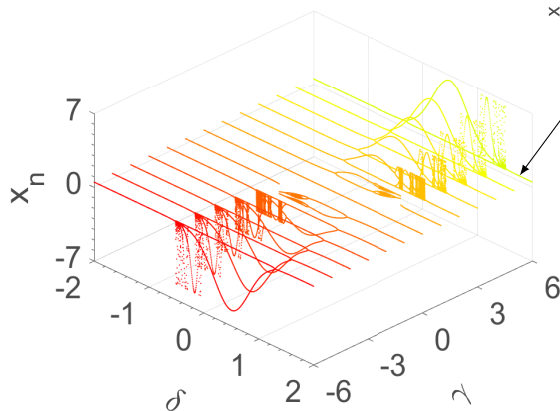


$$\gamma = 5$$

Bifurcation diagrams: population $\times$ mean parameter

$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

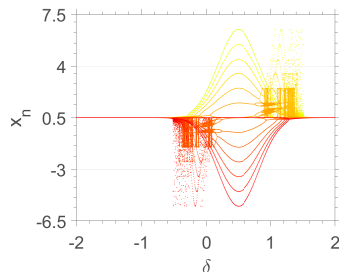
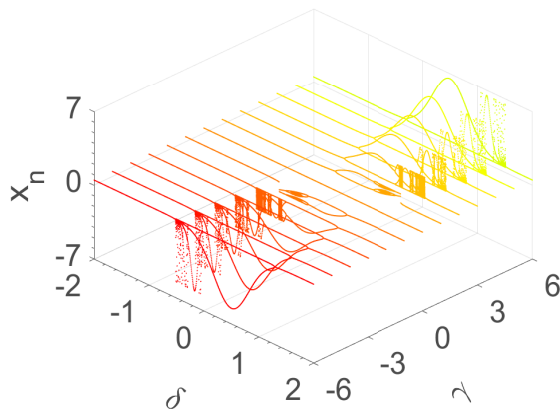


$$\gamma = 6$$

Bifurcation diagrams: population $\times$ mean parameter

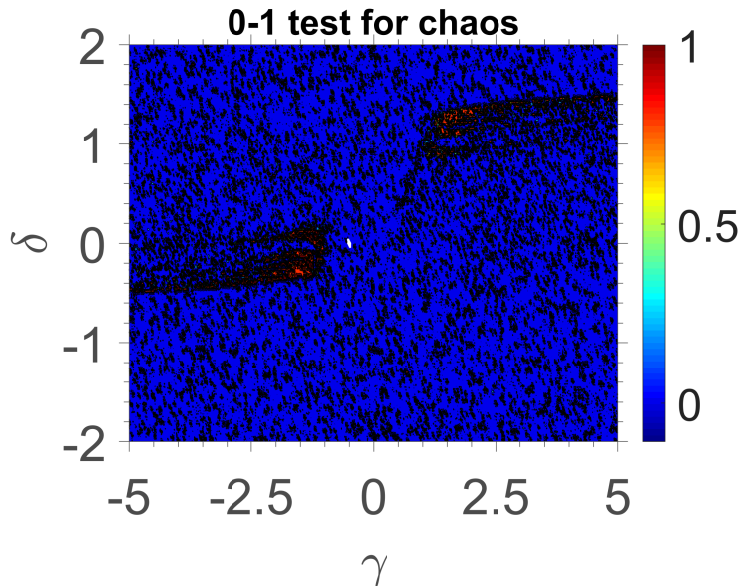
$$\alpha = 4.9 \quad \beta = 0.5$$

$$x_0 = 0.2$$

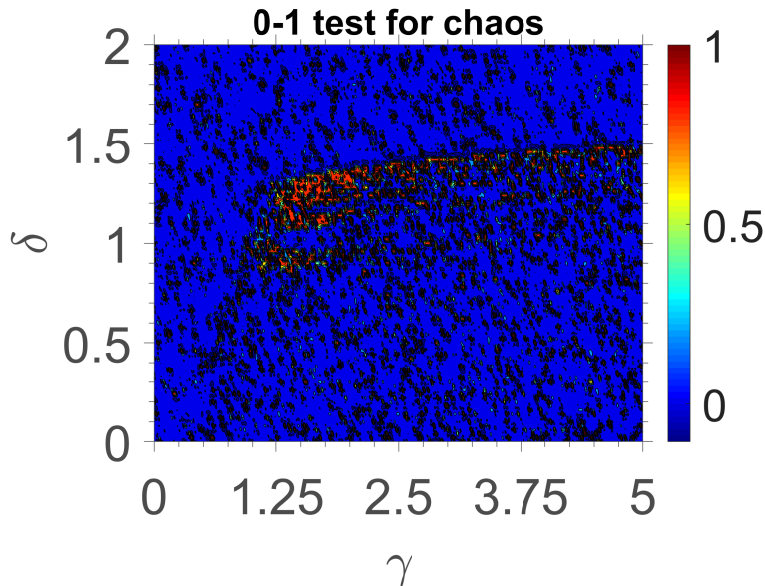


$$\gamma = -6:1:6$$

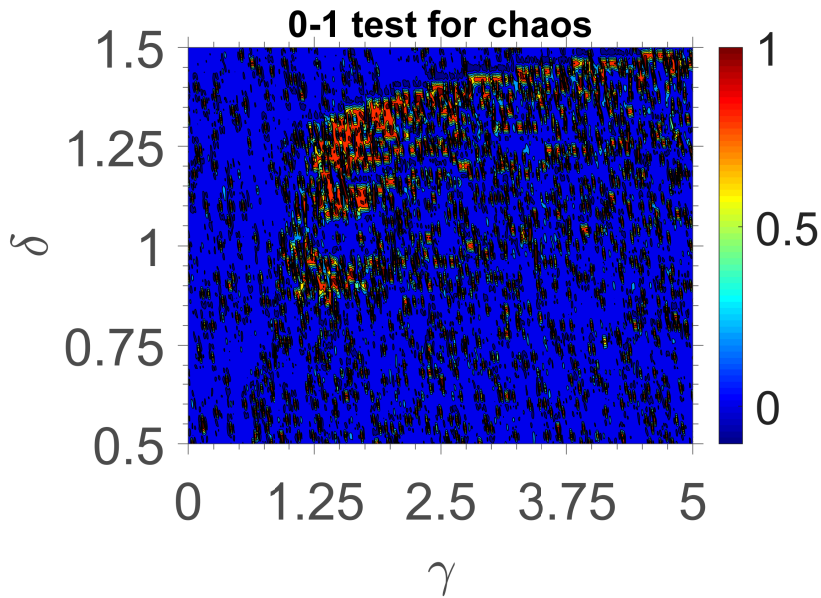
Contour map: scale parameter $\times$ mean parameter



Contour map: scale parameter $\times$ mean parameter



Contour map: scale parameter $\times$ mean parameter



Section 5

Final Remarks

Final remarks

Some conclusions:

- Geometric patters:
 - The bifurcation diagrams in γ from several values of δ have odd symmetry
 - The diagram with $\delta = \beta$ is center of the symmetry
 - The bifurcation diagrams in δ from several values of δ have symmetry with respect to the line $y = \beta$
 - The diagram with $\gamma = 0$ is center of the symmetry
- 0-1 analysis: Contour maps
 - Two principal regions of chaos presence in the plane $\gamma\delta$
 - The odd symmetry can be observed by the contour map
 - In the contour maps are difficulty to determinate the center of the symmetries

Ongoing research:

- Construction of the basins of attraction for the generalized Gauss map

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Thank you for your attention!

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M. Tosin, M.V. Issa, D. Matos, A. do Nascimento and A. Cunha Jr,
Describing the chaotic dynamics of a generalized Gauss iterated map by using the 0-1 test for chaos, 2019
(in preparation).

Steps of 0-1 test for chaos

Given a time series $(\phi_1, \dots, \phi_N)$, for several $c \in (0, 2\pi)$:

- 1 change from coordinates $(\phi, \dot{\phi})$ to (p, q) :

$$p(n) = \sum_{j=1}^n \phi_j \cos(jc), \quad q(n) = \sum_{j=1}^n \phi_j \sin(jc)$$

- 2 compute mean square displacement (for $0 \ll n \ll N$):

$$M(n) = \frac{1}{N} \sum_{j=1}^N \left([p(j+n) - p(j)]^2 + [q(j+n) - q(j)]^2 \right) \\ - \text{mean}(\phi)^2 (1 - \cos(nc)) / (1 - \cos(c))$$

- 3 compute asymptotic growth rate (correlation):

$$K_c = \text{corr} \{ (1, 2, \dots, n), (M(1), M(2), \dots, M(n)) \}$$

- 4 compute median:

$$K = \text{median} \{ K_c \}$$

Numerical classifier:

$K \approx 0$: regular dynamics

$K \approx 1$: chaotic dynamics